

What have we learned so far?

Inferring a phylogeny using the likelihood function



Model for molecular evolution



A mechanism to propose
branch lengths



A mechanism to propose trees

Continuous-Time Markov Chains (CTMC)

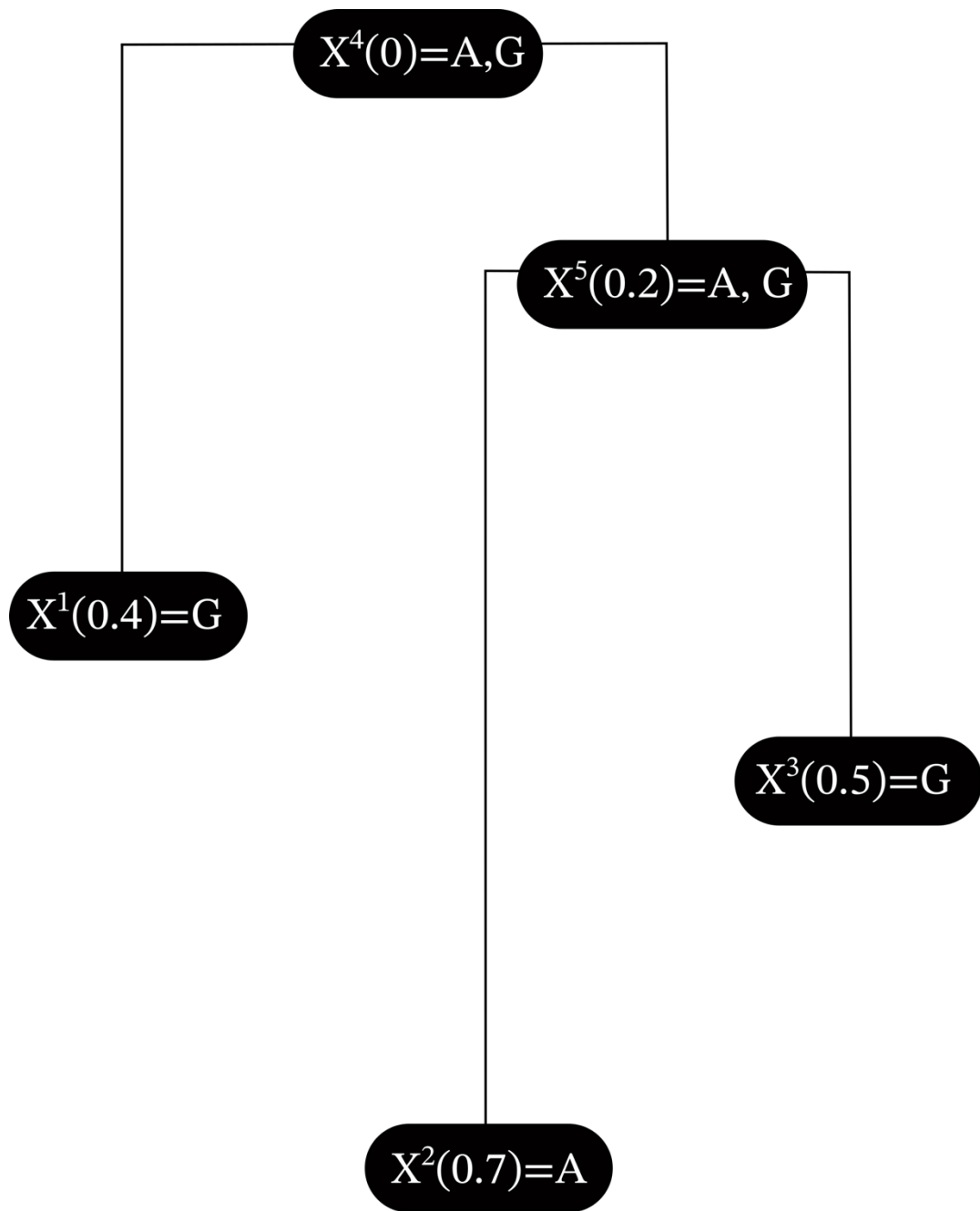
$$\{X(t), t \geq 0\}$$

Stochastic models that follow change in time with an associated **probability**

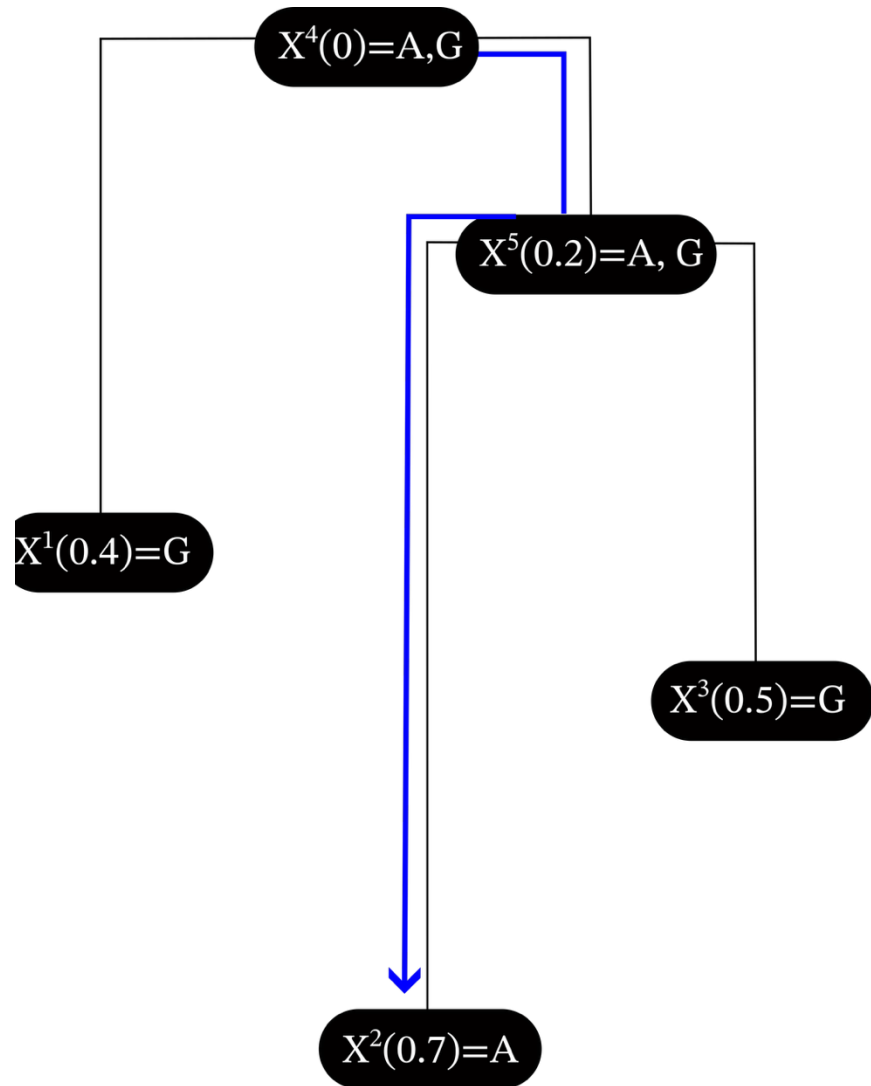
$X(t)$ = Nucleotide at time t

$X(t)$ = A, C, G, T

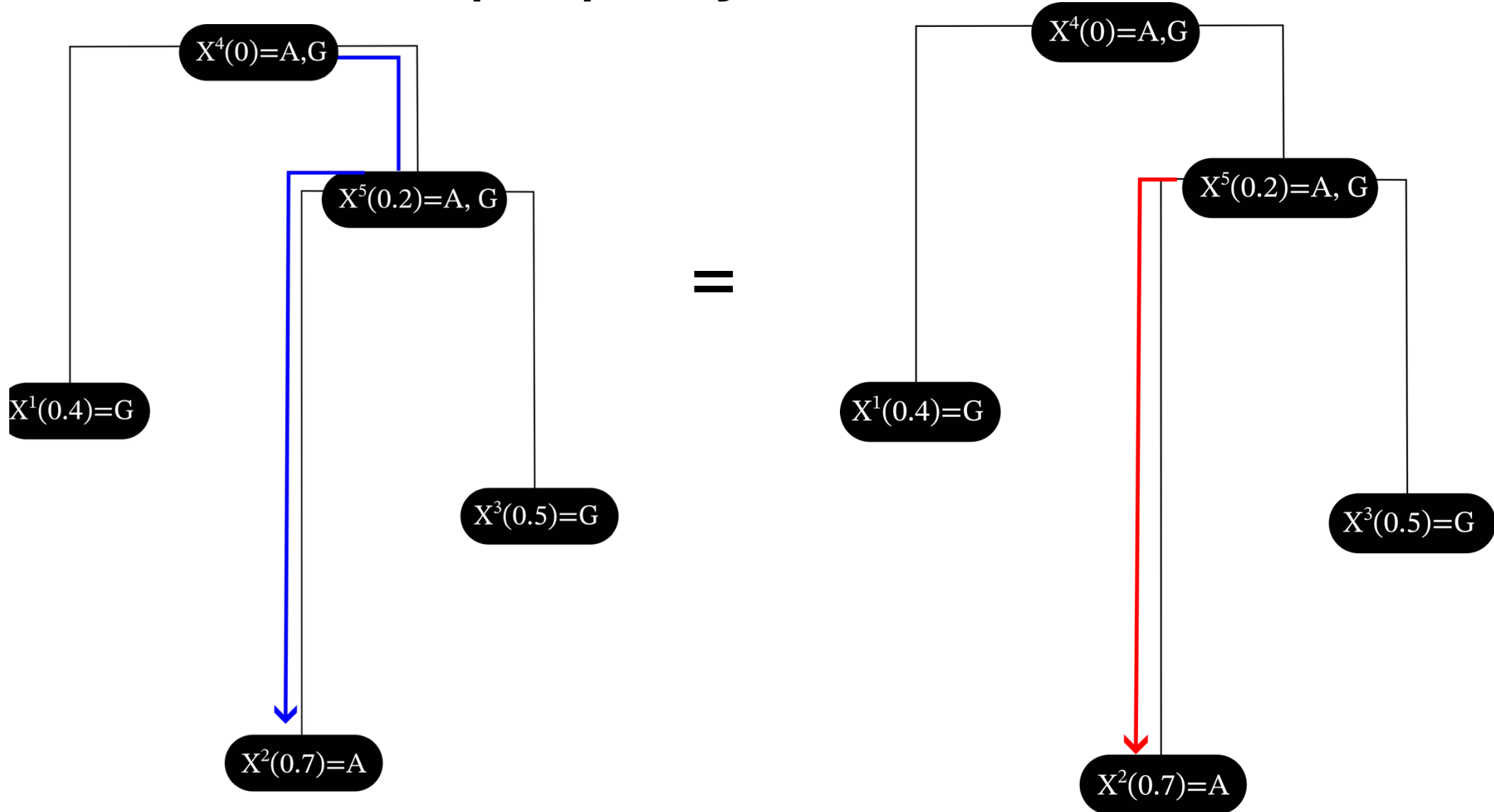
Notation



Getting a nucleotide A in Site 1 for taxon 2

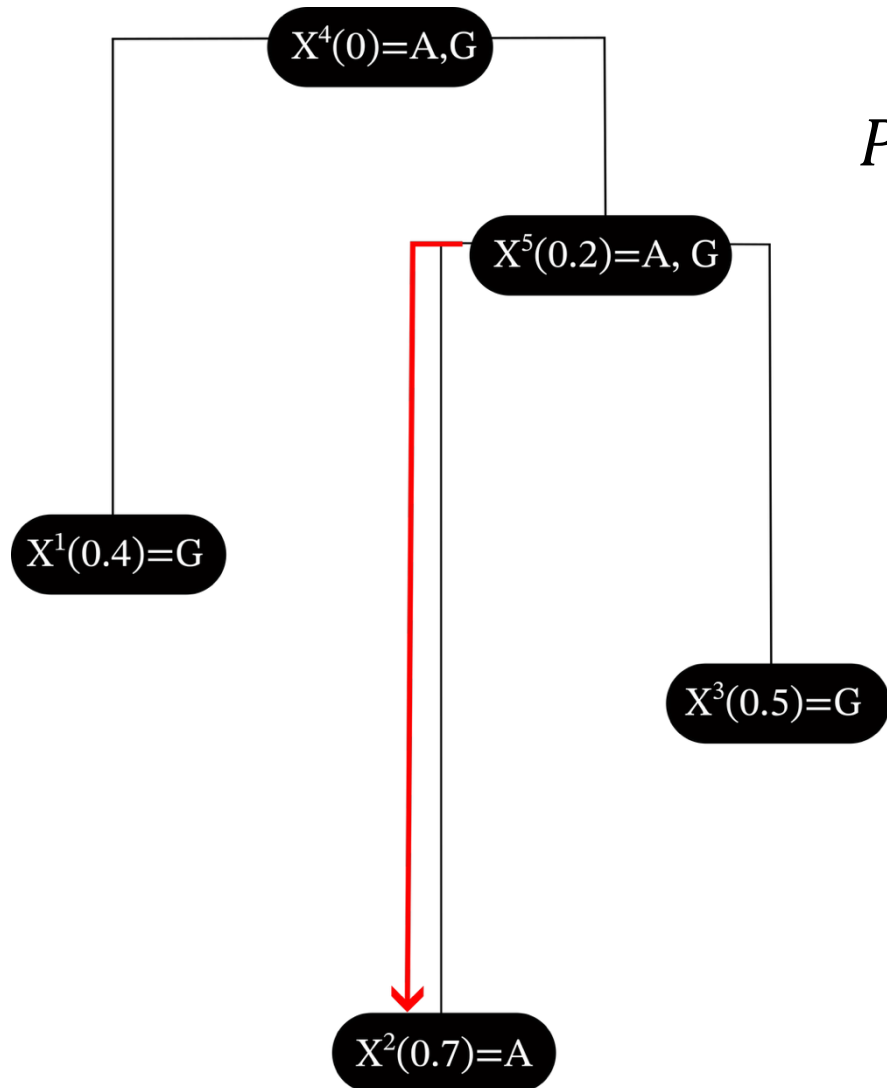


The Markov property



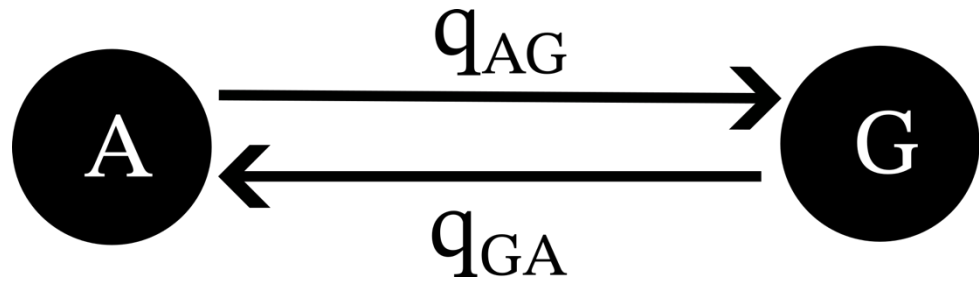
Conditional probabilities

$$P(X^2(0.7)=A|X^5(0.2)=A)=p_{AA}(0.7-0.2)=p_{AA}(0.5))$$



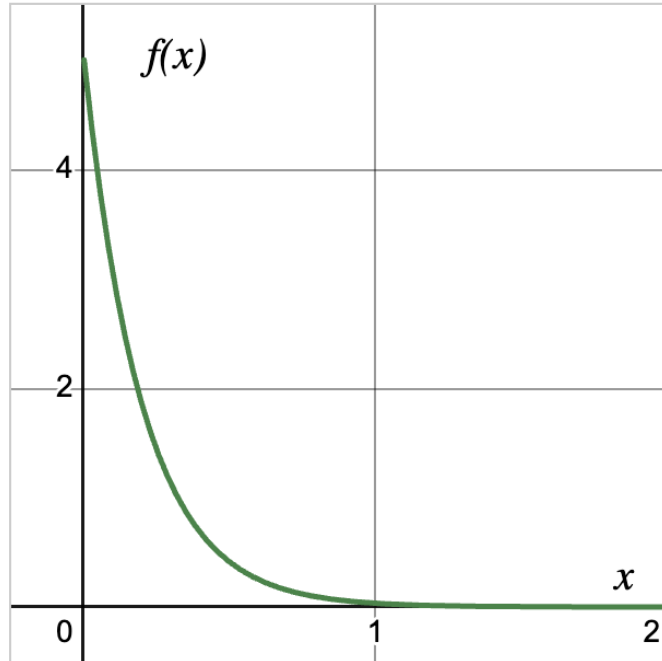
$$P(t) = \begin{pmatrix} p_{AA}(t) & p_{AG}(t) \\ p_{GA}(t) & p_{GG}(t) \end{pmatrix}$$

Transition rates define CTMC



$$\lim_{h \rightarrow 0} \frac{p_{ij}(h)}{h} = q_{ij}$$

Waiting times = Exponential distribution

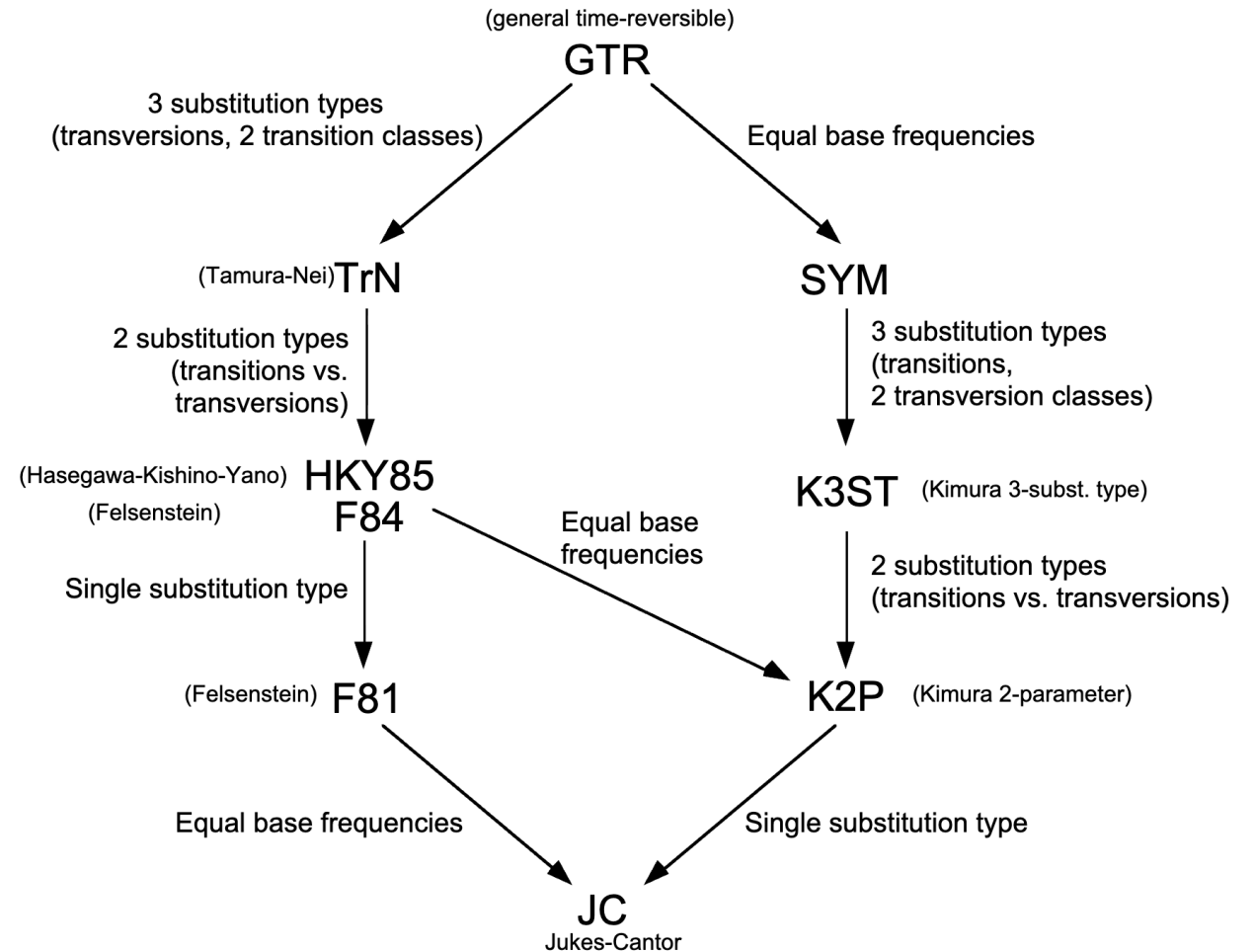


Expected waiting time for G to substitute A is

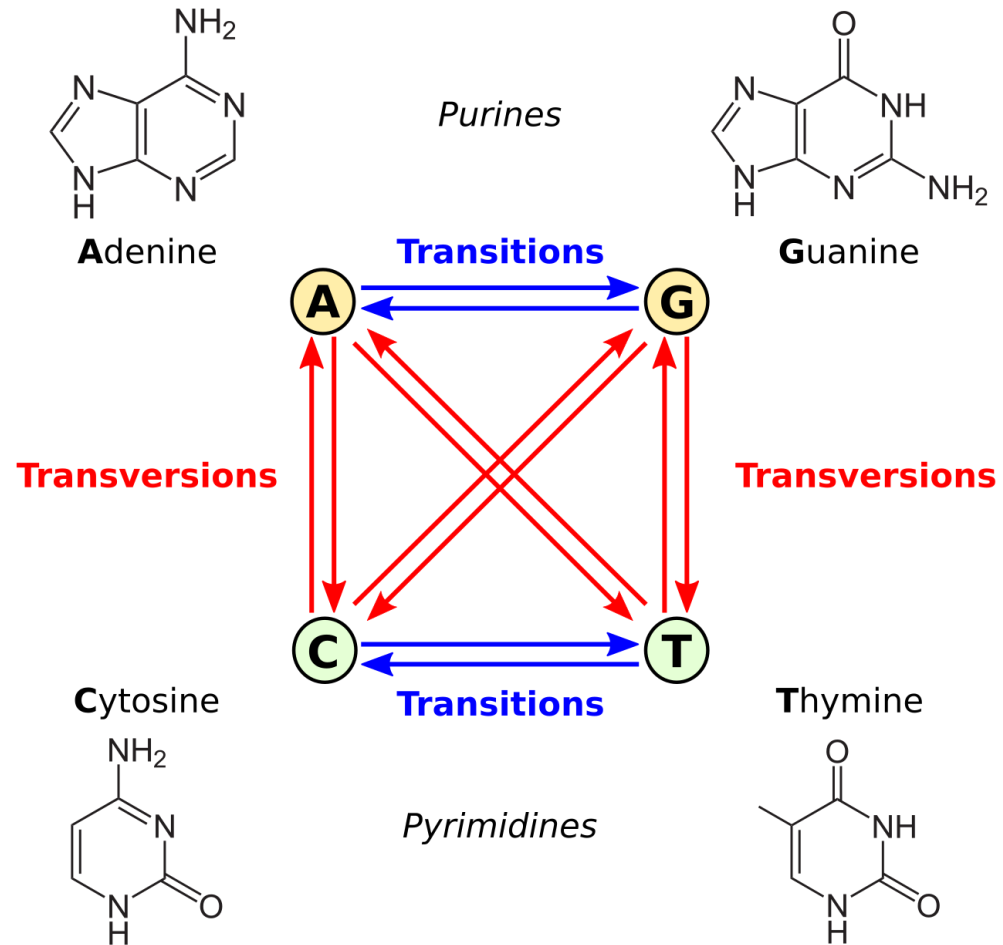
$$\frac{1}{q_{AG}}$$

Models for molecular evolution

GTR Family of Reversible DNA Substitution Models



Substitution rate models



General Time Reversible model (Tavaré, 1986)

$$Q = \begin{pmatrix} -(a\pi_C + b\pi_G + c\pi_T) & a\pi_C & b\pi_G & c\pi_T \\ a\pi_A & -(a\pi_A + d\pi_G + e\pi_T) & d\pi_G & e\pi_T \\ b\pi_A & d\pi_C & -(b\pi_A + d\pi_C + f\pi_T) & f\pi_T \\ c\pi_A & e\pi_C & f\pi_G & -(c\pi_A + e\pi_C + f\pi_G) \end{pmatrix}$$

JC (69)- The simplest nucleotide substitution model

$$a = b = c = d = e = f = \mu$$

$$\pi_A = \pi_C = \pi_G = \pi_T = \frac{1}{4}$$

$$Q = \begin{pmatrix} -(a\pi_C + b\pi_G + c\pi_T) & a\pi_C & b\pi_G & c\pi_T \\ a\pi_A & -(a\pi_A + d\pi_G + e\pi_T) & d\pi_G & e\pi_T \\ b\pi_A & d\pi_C & -(b\pi_A + d\pi_C + f\pi_T) & f\pi_T \\ c\pi_A & e\pi_C & f\pi_G & -(c\pi_A + e\pi_C + f\pi_G) \end{pmatrix}$$

Jukes-Cantor (1969)

The simplest nucleotide substitution model

Transition probabilities
(conditional)

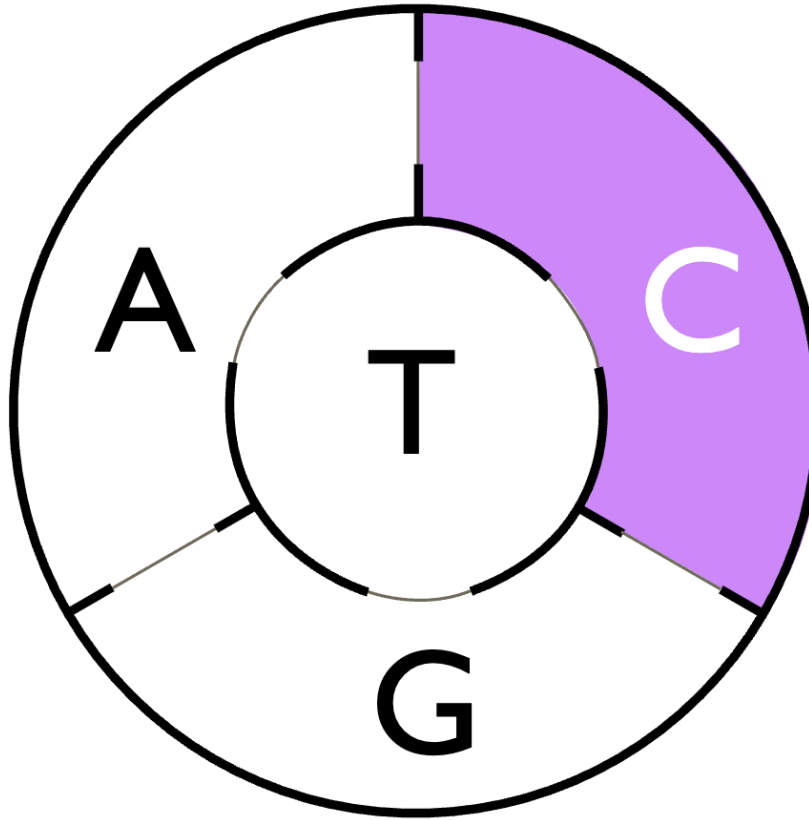
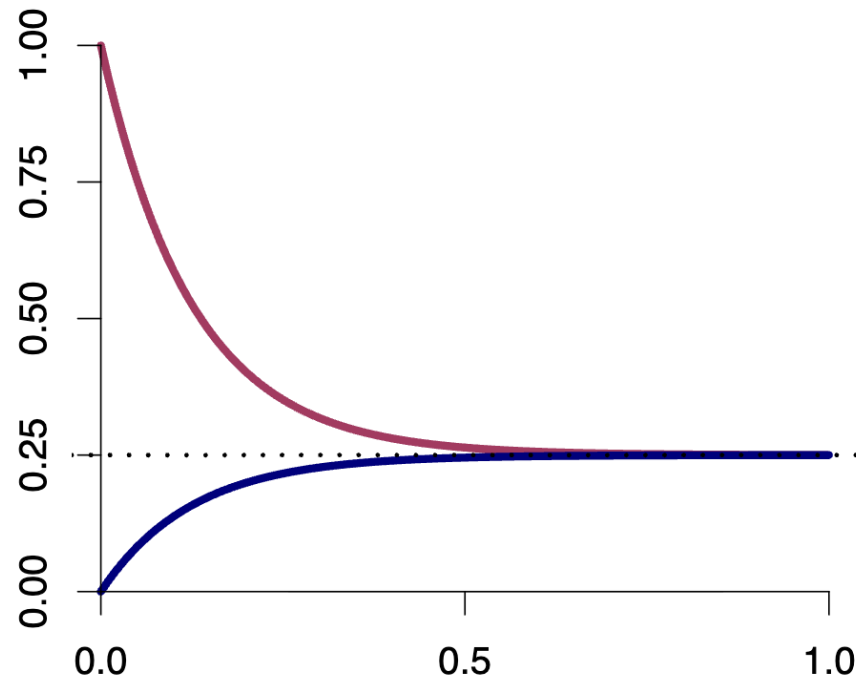
Same but in matrix shape

$$p_{ij}(t) = \begin{cases} \frac{1}{4} + \frac{3}{4}e^{-\mu t} & \text{if } i = j \\ \frac{1}{4} - \frac{1}{4}e^{-\mu t} & \text{if } i \neq j \end{cases}$$

Stationary distribution

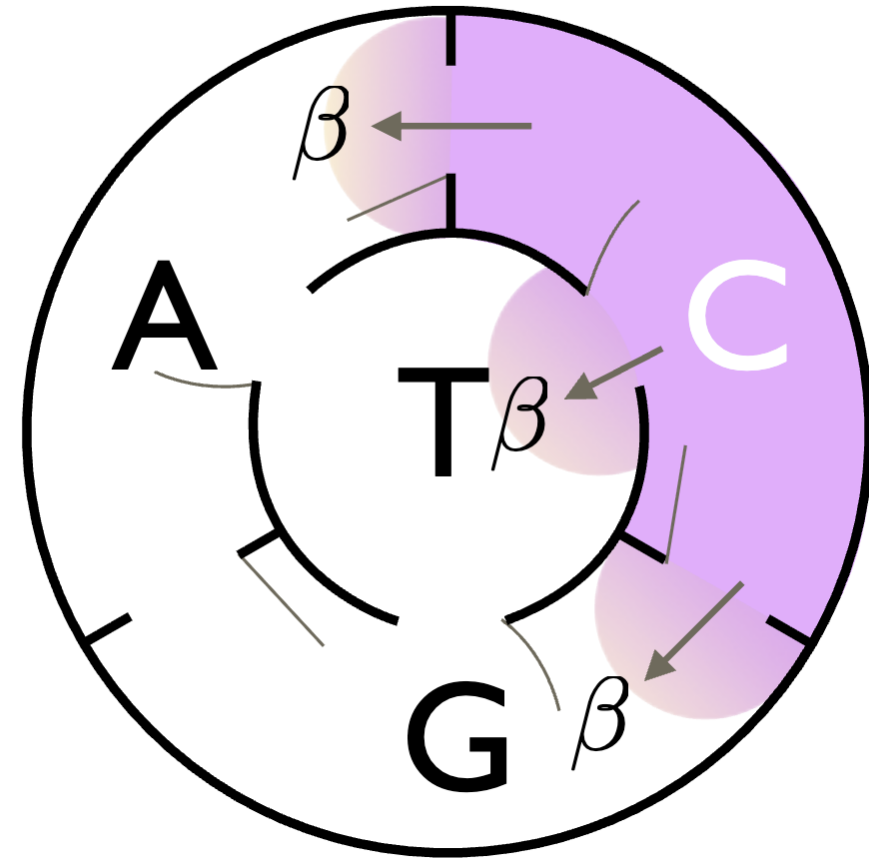
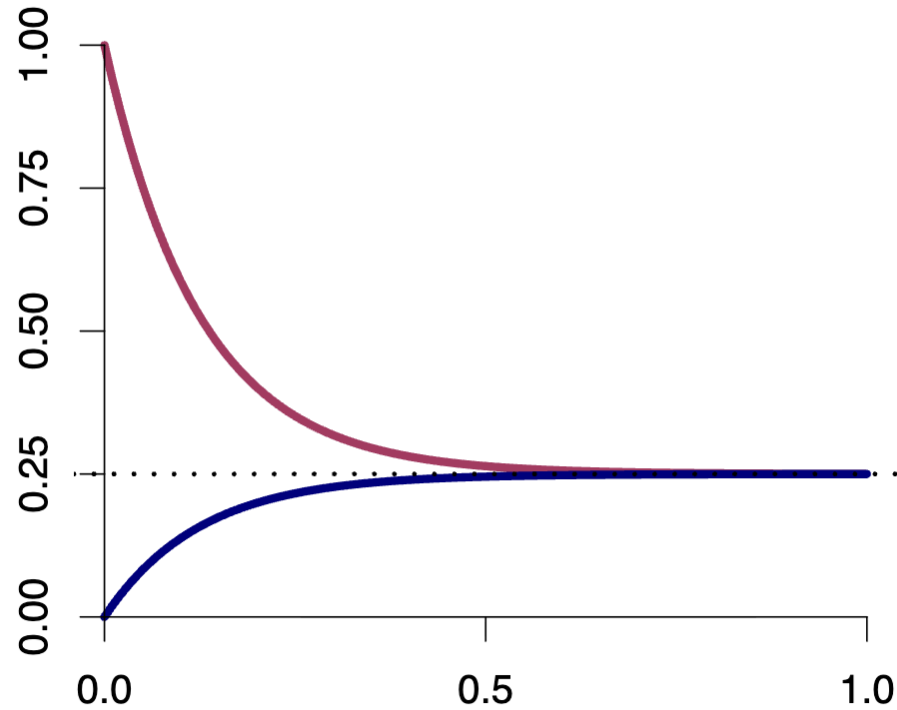
$$\pi \times P = \pi$$

Equilibrium (stationary) probabilities

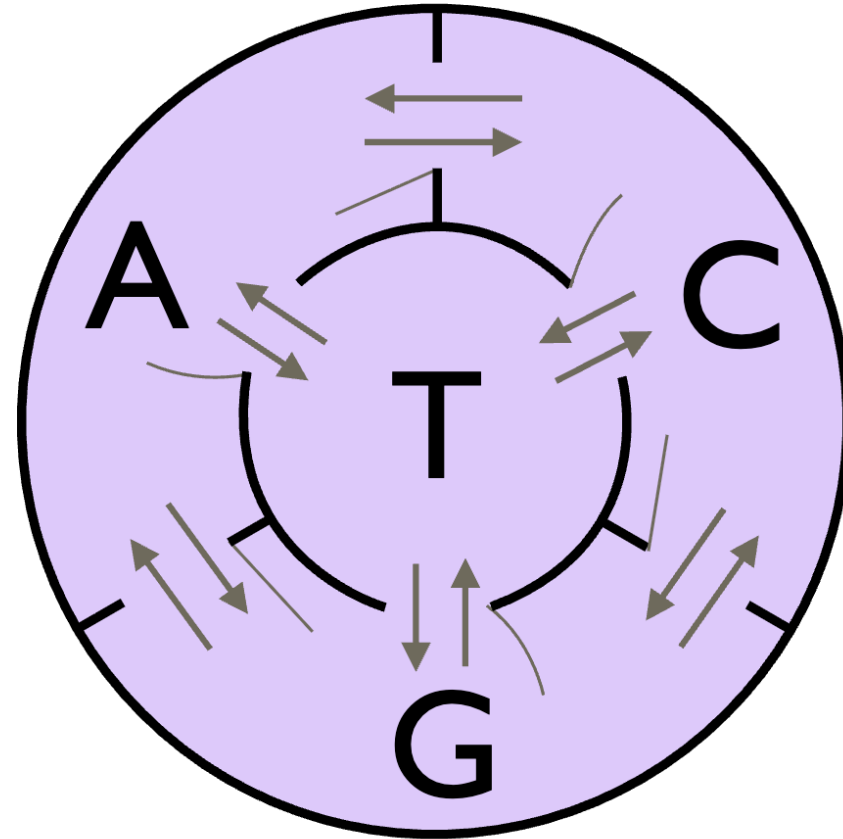
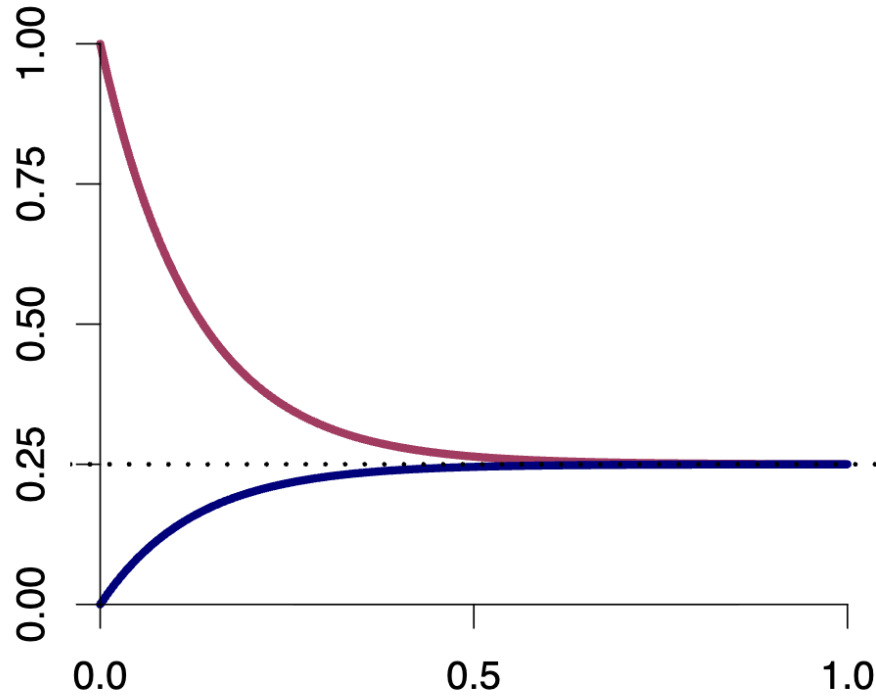


(architect: Joe Bielawski)

Equilibrium (stationary) probabilities



Equilibrium (stationary) probabilities



$$\pi_A = \pi_C = \pi_G = \pi_T = \frac{1}{4}$$

Branches represent expected number of substitutions

$$Q = \begin{pmatrix} -\frac{3}{4}\mu & \frac{1}{4}\mu & \frac{1}{4}\mu & \frac{1}{4}\mu \\ \frac{1}{4}\mu & -\frac{3}{4}\mu & \frac{1}{4}\mu & \frac{1}{4}\mu \\ \frac{1}{4}\mu & \frac{1}{4}\mu & -\frac{3}{4}\mu & \frac{1}{4}\mu \\ \frac{1}{4}\mu & \frac{1}{4}\mu & \frac{1}{4}\mu & -\frac{3}{4}\mu \end{pmatrix}$$

$$p_{ij}(t) = \begin{cases} \frac{1}{4} + \frac{3}{4}e^{-\mu t} & \text{if } i = j \\ \frac{1}{4} - \frac{1}{4}e^{-\mu t} & \text{if } i \neq j \end{cases}$$

Example: Evolutionary distance under JC(69)

- Taxon 1: A A C G T A C G A A
- Taxon 2: A A T G T A C A A A

Proportion of differences $z=0.2$

Exercise: Building new models of substitution rates.

After you write the model tell me, what do you think the rates represent?

$$Q = \begin{pmatrix} -(a\pi_C + b\pi_G + c\pi_T) & a\pi_C & b\pi_G & c\pi_T \\ a\pi_A & -(a\pi_A + d\pi_G + e\pi_T) & d\pi_G & e\pi_T \\ b\pi_A & d\pi_C & -(b\pi_A + d\pi_C + f\pi_T) & f\pi_T \\ c\pi_A & e\pi_C & f\pi_G & -(c\pi_A + e\pi_C + f\pi_G) \end{pmatrix}$$

Group 1

$$a = b = c = d = e = f = \mu$$

(you can make $\mu = 1$)

$(\pi_A, \pi_C, \pi_G, \pi_T)$ all free

Group 2

$$a = c = d = f = \mu$$

$$b = e = \mu\kappa$$

(you can make $\mu = 1$)

$$\pi_A = \pi_C = \pi_G = \pi_T = \frac{1}{4}$$

Group 3

$$a = c = d = f = \mu$$

$$b = e = \mu\kappa$$

(you can make $\mu = 1$)

$(\pi_A, \pi_C, \pi_G, \pi_T)$ all free

Group 4

$$a = c = d = f = \mu$$

$$b = \mu\kappa_1$$

$$e = \mu\kappa_2$$

$(\pi_A, \pi_C, \pi_G, \pi_T)$ all free

Exercise: Writing your own model of molecular evolution

Group 1

F81

Felsenstein 1981

Group 2

K2P

Kimura 1980

Group 3

HKY85

Hasegawa et al. 1985

Group 4

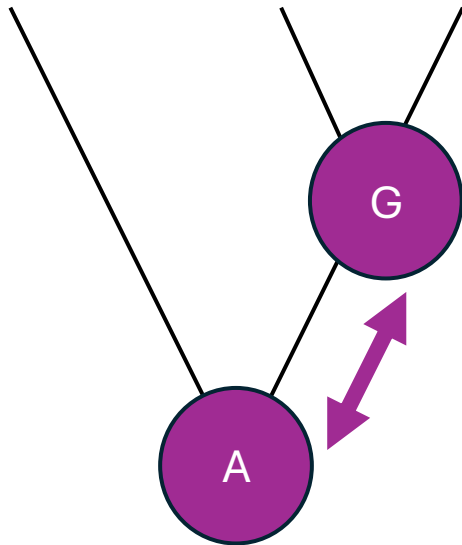
TN93

Tamura and Nei (1993)

A note on the reversible adjective

The reversibility property

$$\pi_A p_{AG}(t) = \pi_G p_{GA}(t)$$



- Any point of the tree can serve as the root
- Time reversibility is not necessary for substitution models

t1	G	C	T	T	C	T	G	A	T	T	A	A	C	C	T	G	C	T
t2	G	C	T	T	C	T	G	A	T	T	T	C	T	C	T	G	C	C
t3	G	C	T	T	C	T	G	A	T	T	A	C	T	C	T	G	C	C
t4	G	C	T	T	C	T	G	A	C	T	A	G	T	C	T	G	C	T

- Invariant sites (I)
- Gamma (G)

$$r_1 P_1(t) + r_2 P_2(t) + \cdots + r_k P_k(t)$$



Analogy from Paul Lewis about a model with varying substitution rates (illustrated by ChatGPT)

You buy a bunch of different shirt sizes (substitution rates) for a group of different women (sites).

It will be more expensive than buying a one-size fits all. But women will be happier with their own size rather than one-size fits all.